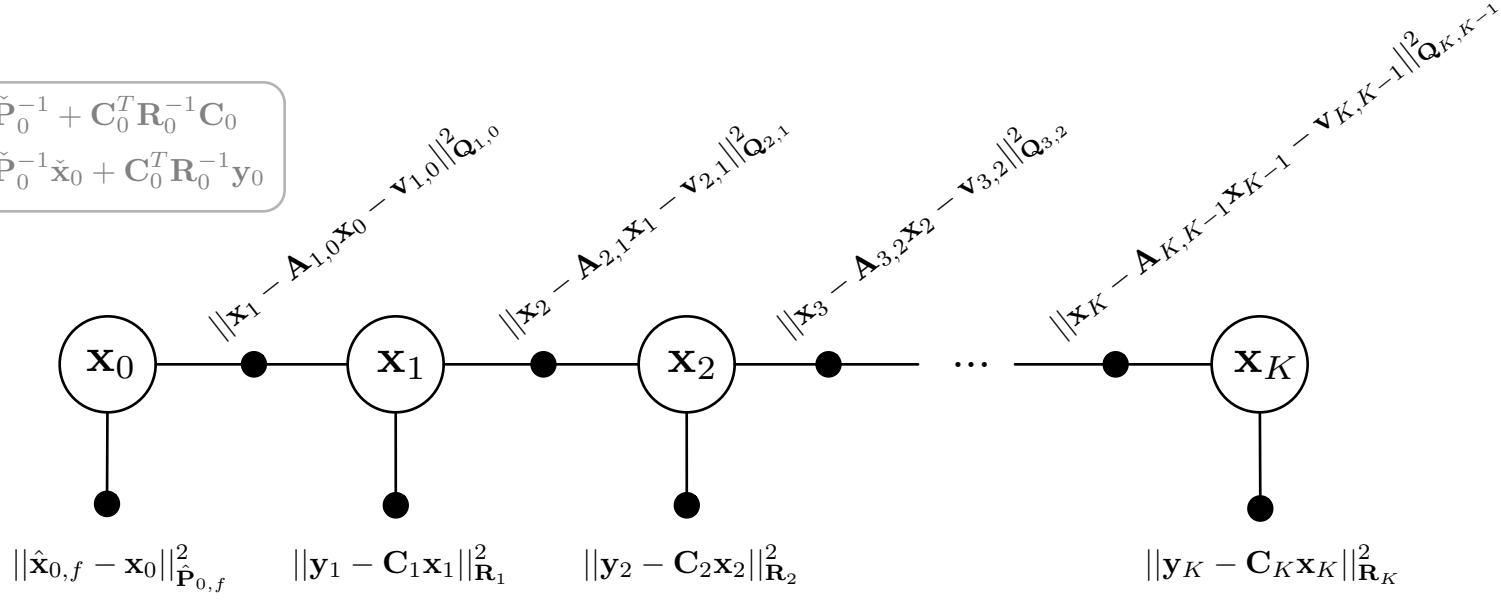$$\hat{\mathbf{P}}_{0,f}^{-1} = \check{\mathbf{P}}_0^{-1} + \mathbf{C}_0^T \mathbf{R}_0^{-1} \mathbf{C}_0$$
$$\hat{\mathbf{P}}_{0,f}^{-1} \hat{\mathbf{x}}_{0,f} = \check{\mathbf{P}}_0^{-1} \check{\mathbf{x}}_0 + \mathbf{C}_0^T \mathbf{R}_0^{-1} \mathbf{y}_0$$